

Procedure to Execute Assembly Language Program 8051 using IDE Simulator.

1. Select 8051 IDE Tool on Desktop.
2. Select file. Select New.
3. Write Assembly language program.
4. Save Program
5. Select Assembler on task bar.
6. Select Simulate, Start Simulator.
7. Execute program, Click on step into.
8. Continue on step into, up to end of blue arrow.
9. See the result on view tab.(Register, External memory.
internal memory)

Procedure to use Keil uVision 5 (8051)

- ① Create folder on D Drive
- ② Open Keil Micro Vision .5
- ③ Click on project menu
- ④ select New micro vision project
- ⑤ Give file name to project and save it (D:\folder\filename)
- ⑥ select device from open window (AT89C51)
- ⑦ click OK
- ⑧ click No on open window
- * ⑨ Create new file, click on new file
- ⑩ Type program
- ⑪ Save program with your file name given already with .asm
- * ⑫ Add file to source group 1 (under project → target)
 - ① Right click on source group 1 (Drop down of project)
 - ② click on add new file / existing file
 - ③ Select Drive, select folder, select file (Asm file source .asm)
select file to add
 - ④ click on Add ⑤ click on close.
- * Assemble, compile (build) file
 - ⑬ click on translate (shortcut key) (Ctrl+F7) - (to assemble)
 - ⑭ click on build, rebuild.
 - ⑮ click on Debug, click on start/stop debug session
 - ⑯ click on OK (evaluation mode)
 - ⑰ click on debug, click on Run (execute program)
- * To see result :-
 - * check content of registers displayed on open window
 - * check result of port - click peripheral, click ports
 - * check result of memory - click on view, click memory window
Type Address: d: memory location ex: d:20h
- * Create .hex file to store on 8051 chip :-
 - ① ^{Right} click on Target (Drop down of project)
 - ② click on option for target 1
 - ③ Type xtal(mhz) = 11.0592 in target tab, Device AT89C51
 - ④ click OK.
 - ⑤ click on shortcut keys, translate, build, rebuild
 - ⑥ check hex file in your folder, open object folder, see hex file.

Steps to Burn Program in 8051 Microcontroller

1. Connect USB programmer to PC
2. Insert 89c51 Controller on USB kit.
3. Open folder → CD for 8051 development (On PC)
4. Open Folder – 8051 USB Programmer CD
5. Open – WIPRO software *(Flash magic)*
6. Click on LOAD
7. Click on Hex file
8. Open Hex file
9. Click OK
10. Click Auto
11. Remove 89c51 from USB kit
12. Place 89c 51 on Target System
13. Connect power supply to target System
14. See the result on Target System.